Worksheet - Stellar Model

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1 Gravitational Force

(a) Recap

i) What is the gravitational force between a planet of mass M_1 and a star of mass M_2 separated by a distance R?

ii) How many times would the force increase if the distance were halved?

(b) Mass in a sphere

Imagine you have a sphere of mass M and radius R, which has uniform density throughout. If you choose a smaller part of this sphere with radius r, what is the mass of this smaller sphere $M_s(r)$?

(c) Force Now let's say in this sphere of cookie dough, I have a raisin at a distance r from the center. What is the gravitational force on this raisin, if it has m_r mass?

2 Ideal Gas Law

I have an ideal gas at temperature T with particle density $n = N/V$, where N is total number of particles and V is volume. What is the pressure of the gas in terms of n and Boltzmann constant and temperature?

Optional: Remember what conditions make a gas follow the ideal gas law. Do you think this should be applicable to stars?

3 Radiation Pressure - Introduction

(a) Planck's Law

i) Planck's equation of quantization of energy states that light is made of particles called photons, and each photon carries an energy determined by the frequency of the light.

$$E = h\nu \tag{1}$$

Here $h = 6.6 \times 10^{-34} J.s$ is Planck's constant.

Using this equation, calculate the energy of a photon of a light with frequency $\nu = 1000$ Hz.

ii) If a wavelength of a light is λ, then the relationship between the wavelength and the frequency is $\lambda\nu = c$, where $c = 3 \times 10^8\ m/s$ is the speed of light.

Using this equation and Planck's equation, calculate the energy of a photon of red light (wavelength = 700 nm) and violet light (wavelength = 400 nm).

(b) Pressure from radiation

A light of frequency ν Hz and intensity I Wm^{-2} is incident on a perfectly reflective plate. What is the radiation pressure on the plate?

(c) Radiation Pressure Law

The radiation pressure of a star at temperature T is given by

$$P_R = \frac{1}{3} a T^4 \tag{2}$$

$a = 7.6 \times 10^{-16}\ Jm^{-3}K^{-4}$.

Calculate the radiation pressure in the sun ($T = 5000K$) and in a typical neutron star ($T = 10^{12}K$).

4 Hydrostatic Equilibrium

(a) Introduction

i) Suppose you have a glass of water, with area = A and the height the water reaches is H_w from the bottom of the glass. Consider a slab of this water at a height h, the thickness of the slab being Δh. Density of water is ρ.

What are the forces on this slab of water? Draw the force diagram.

ii) Using the fact that this slab is stationary, derive a relationship between the pressure difference ($P_{\text{up}} - P_{\text{down}} = \Delta P$) and the thickness of the slab Δh.

(Hint: Calculate the mass of the slab using the density and the dimensions given)

(b) Hydrostatic Equilibrium in a star

Step I: Consider a thin shell in the star (mass M, radius R), which has radius r and thickness Δr. What are the forces on this shell? Draw the force diagram.

Step II: Now using the fact that this shell is stationary, derive a relationship between the pressure difference ($P_{\text{out}} - P_{\text{in}} = \Delta P$) and the thickness of the shell Δr.

Step III: Calculate $\Delta P/\Delta r$. When both ΔP and Δr are infinitesimal, this equation can be written in terms of the dP/dr.

Step IV: Now substitute $P = P_R + P_g$, where P_R and P_g are the radiation pressure and gas pressure respectively. Substitute the values of these pressures that you found in the earlier parts of this worksheet.

And we officially have the hydrostatic equilibrium equation of a star!!